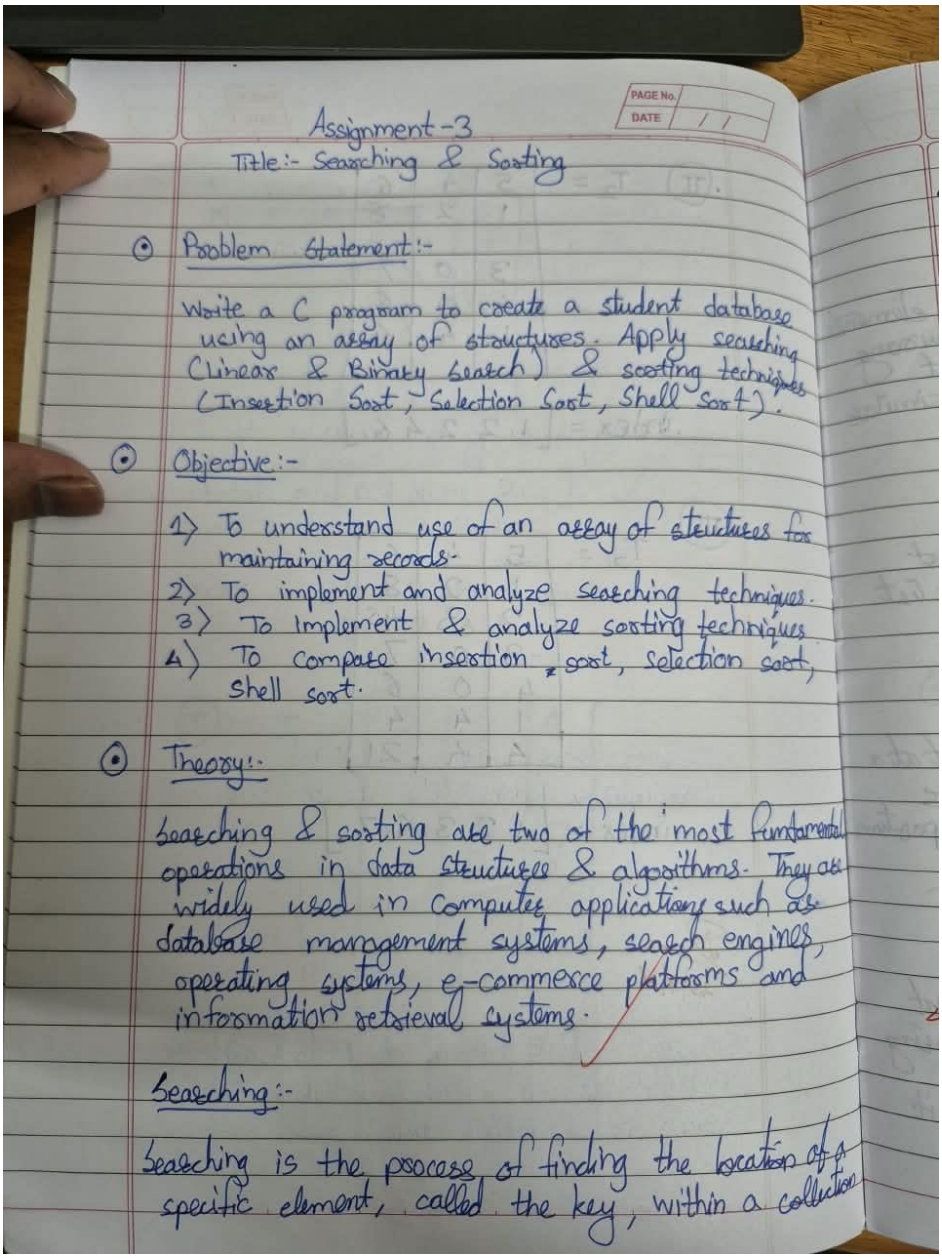
ASSIGNMENT-3

NAME – Abhinav Ajitkumar Chavan

PRN – 1262251966

BATCH - A1



Assignment-3

Title:- Searching & Sorting

⊙ Problem Statement:-

Write a C program to create a student database using an array of structures. Apply searching (Linear & Binary Search) & sorting techniques (Insertion Sort, Selection Sort, Shell Sort).

⊙ Objective:-

1) To understand use of an array of structures for maintaining records.
2) To implement and analyze searching techniques.
3) To implement & analyze sorting techniques.
4) To compare insertion sort, selection sort, shell sort.

⊙ Theory:-

Searching & sorting are two of the most fundamental operations in data structures & algorithms. They are widely used in computer applications such as database management systems, search engines, operating systems, e-commerce platforms and information retrieval systems.

Searching:-

Searching is the process of finding the location of a specific element, called the key, within a collection

of data. If the element is present, the search operation returns its position; otherwise, it indicates that the element is not found.

Types of Searching:-

1) Linear Search:-

Linear Search, also known as Sequential Search, is the simplest searching technique. In this method, each element of the list is compared one by one with the target element until a match is found or the entire list has been searched.

2) Binary Search:-

Binary Search is an efficient searching algorithm that works only on sorted data. It repeatedly divides the search interval into two halves until the target element is found or the search space becomes empty.

Sorting:-

Sorting is the process of arranging data elements in a specific order, usually ascending or descending. Sorted data allows faster searching, easier analysis & efficient data processing.

PAGE No.
DATE / /

Types of Sorting :-

1) Bubble Sort

Bubble Sort repeatedly compares adjacent elements & swaps them if they are in the wrong order. After each pass, the largest element moves to its correct position, similar to bubbles rising to the surface.

2) Selection Sort

Selection Sort repeatedly selects the smallest element from the unsorted portion of the list & places it at beginning.

3) Insertion Sort

Insertion Sort builds a sorted list one element at a time by inserting each new element into its correct position in the already sorted portion of the array.

4) Shell Sort

Shell Sort allows the exchange of elements that are far apart. It sorts elements by comparing & swapping items separated by a specific gap, gradually reducing the gap until it becomes 1.

5) Merge Sort

Merge Sort is a divide-and-conquer algorithm that divides the array into smaller subarrays, sorts each subarray recursively, and then merges them into a sorted array.

6) Quick Sort

Quick sort is another divide-and-conquer algorithm. It selects a pivot element & partitions the array into two parts such that elements smaller than the pivot are placed before it and larger elements after it. The process is repeated recursively.

④ Implementation:-

- Platform:-

→ 64-bit Open Source Linux or its derivatives

→ Open Source C Programming tool like gcc / Eclipse Editor.

⑤ Test Conditions:-

1) Input at least five records.
2) Search a student in the record as per the roll number. (Linear search).

PAGE No.
DATE / /

3) Arrange list of students roll call listwise.
4) Search a student as per roll number in the sorted list.
5) Test the searching & sorting methods for at least five records.

⊙ Conclusion:-

Thus, implemented different searching & sorting methods on the student database. This system is able to perform sorting under different cases.

⊙ FAQs:-

1) What is the meaning of database? How to maintain in C.

Ans) Database:-

A database is an organized collection of related data that can be stored, retrieved, updated & managed efficiently.

Maintaining a Database in C:-

- Structures to store records
- Arrays of structures
- Files (.txt or .dat) for permanent storage

2) What are the applications of searching?

Ans)
- Searching contacts in a phone.
- Searching products on Amazon.
- Searching books in a library.
- Searching files in a computer.

3) Compare linear search & binary search?

	Feature	Linear Search	Binary Search
i)	Technique	Sequential search	Divide & conquer
ii)	Data requirement	Can work on unsorted data	Requires sorted data
iii)	Time Complexity	O(n)	O(log n)
iv)	Best case	O(1)	O(1)
v)	Worst case	O(n)	O(log n)
vi)	Implementation	Very simple	Slightly more complex

4) What are the applications of sorting?

Ans)
- Faster searching
- Database management
- Library management systems
- Student result ranking

PAGE No.
DATE / /

5) Compare Selection Sort, Insertion Sort, & Shell Sort.

	Feature	Selection Sort	Insertion Sort	Shell Sort
(i)	Working	Select minimum element & place it correctly	Insert each element into its correct position.	Sort elements with decreasing gaps.
(ii)	Best Case	$O(n^2)$	$O(n)$	Around $O(n \log n)$
(iii)	Average Case	$O(n^2)$	$O(n^2)$	Approximately $O(n^{1.3} - n^{1.5})$
(iv)	Worst Case	$O(n^2)$	$O(n^2)$	$O(n^2)$

6) What are External & Internal Sorting, In-place Sorting & Stable Sorting ?

Ans)

A] Internal Sorting :-

Sorting performed when all data fits into the main memory (RAM).

B] External Sorting :-

Sorting performed when the data is too large to fit into RAM & must use secondary storage.

C.] In-place Sorting :-

An algorithm is in-place if it uses only a small, constant amount of extra memory while sorting.

D] Stable Sorting :-

An sorting algorithm is stable if equal elements keep their original relative order after sorting.

⊙ Pseudo Code :-

```
Start
Input n

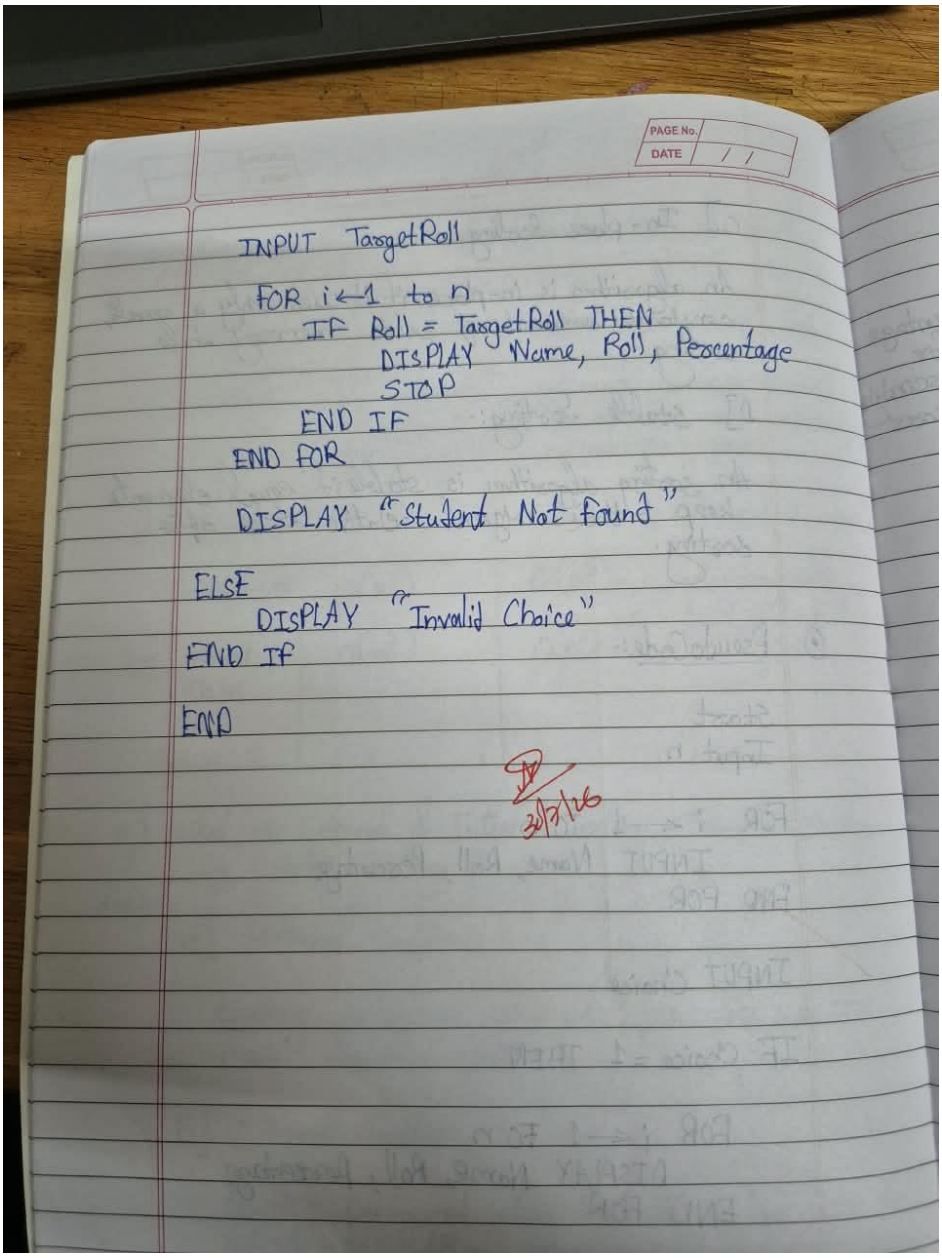
FOR i ← 1 TO n
    INPUT Name, Roll, Percentage
END FOR

INPUT Choice

IF Choice = 1 THEN

    FOR i ← 1 TO n
        DISPLAY Name, Roll, Percentage
    END FOR

ELSE IF Choice = 2 THEN
```



PAGE No.
DATE / /

```
INPUT TargetRoll

FOR i←1 to n
    IF Roll = TargetRoll THEN
        DISPLAY Name, Roll, Percentage
        STOP
    END IF
END FOR

DISPLAY "Student Not found"

ELSE
    DISPLAY "Invalid Choice"
END IF

END
```

30/3/26